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MEMORY MANAGEMENT

Main memory (RAM) is an important resource that must be very carefully managed. While the average home computer nowadays has 100,000 times more memory than the IBM 7094, the largest computer in the world in the early 1960s, programs are getting bigger faster than memories. To paraphrase Parkinson's Law, "Programs expand to fill the memory available to hold them." In this chapter, we will study how operating systems create abstractions from memory and how they manage them.

What every programmer would like is a private, infinitely large, infinitely fast memory that is also nonvolatile, that is, does not lose its contents when the electric power is switched off. While we are at it, why not make it inexpensive, too? Unfortunately, technology does not provide such memories at present. Maybe you will discover how to do it.

What is the second choice? Over the years, people discovered the concept of a **memory hierarchy**, in which computers have a few megabytes of very fast, expensive, volatile cache memory, a few gigabytes of medium-speed, medium-priced, volatile main memory, and a few terabytes of slow, cheap, nonvolatile magnetic or solid-state storage, not to mention removable storage, such as USB sticks. It is the job of the operating system to abstract this hierarchy into a useful model and then manage the abstraction.

The part of the operating system that manages (part of) the memory hierarchy is called the **memory manager**. Its job is to efficiently manage memory: keep track of which parts of memory are in use, allocate memory to processes when they need it, and deallocate it when they are done.